UET.CS2043 - Lập trình nâng cao

Concurrency & Multi-threaded Programming

Contents

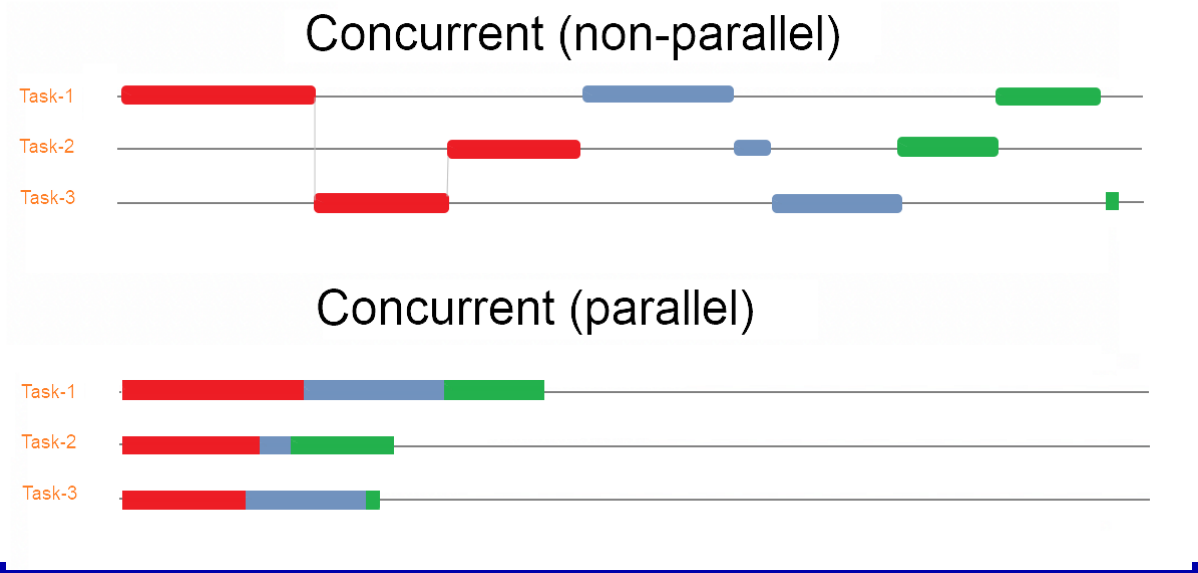
- Concurrency and Multithreading
- Thread and Runnable in Java
- Executor Service & Thread Pool
- Callable and Future
- Thread Synchronization
- Locks and Atomic Variables
- Virtual Thread

“Why” of Concurrency

Core concepts

Concurrency

- Concurrency: run several programs or several parts of a program in parallel
- A task can be performed asynchronously or in parallel
  - → Improves the throughput and the interactivity of the program.

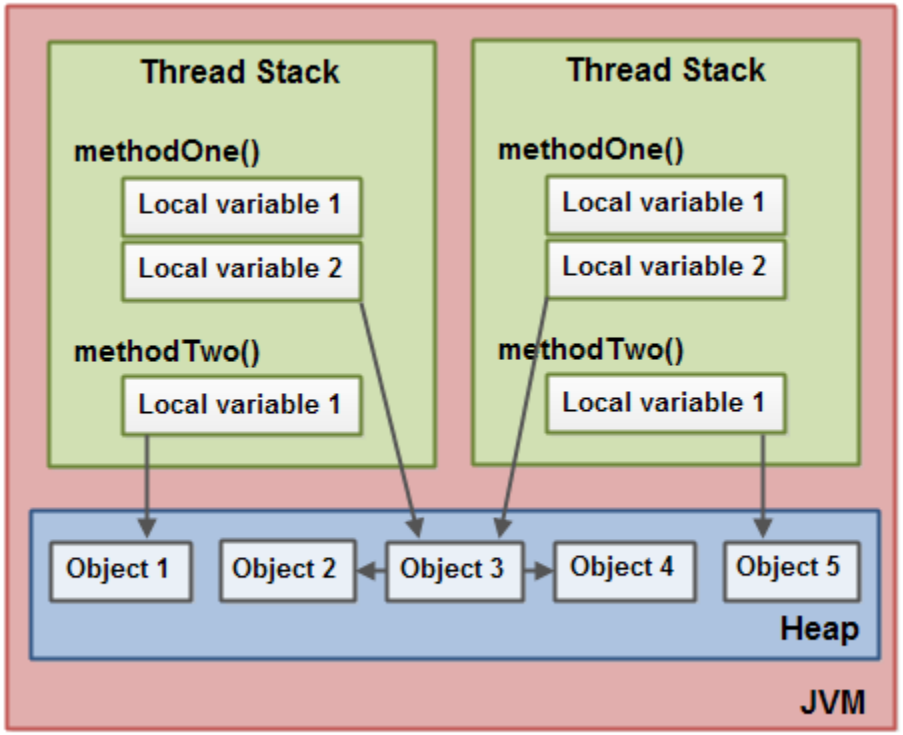


Unit of Concurrency

- Multi-processing - Multiple Processors/CPUs executing concurrently (Unit: CPU)
- Multi-tasking - Multiple tasks/processes running concurrently on a single CPU.
  - OS executes these tasks by switching between them very frequently (Unit: Process)
- Multi-threading - Multiple parts of the same program running concurrently.
  - Dividing the same program into multiple parts/threads and run those threads concurrently (Unit: Thread)

Processes and Threads

- A process is a program in execution running independently and isolated from others
- A thread is a path of execution within a process
  - It has its own call stack but can access shared data of other threads in the same process.
- A Java application runs by default in one process
- Within a Java application might have several threads to achieve parallel processing or asynchronous behavior


Thread Stack
methodOne()
Local variable 1
Local variable 2
methodTwo()
Local variable 1
Thread Stack
methodOne()
Local variable 1
Local variable 2
methodTwo()
Local variable 1
Object 1
Object 2
Object 3
Object 4
Object 5
Heap
JVM

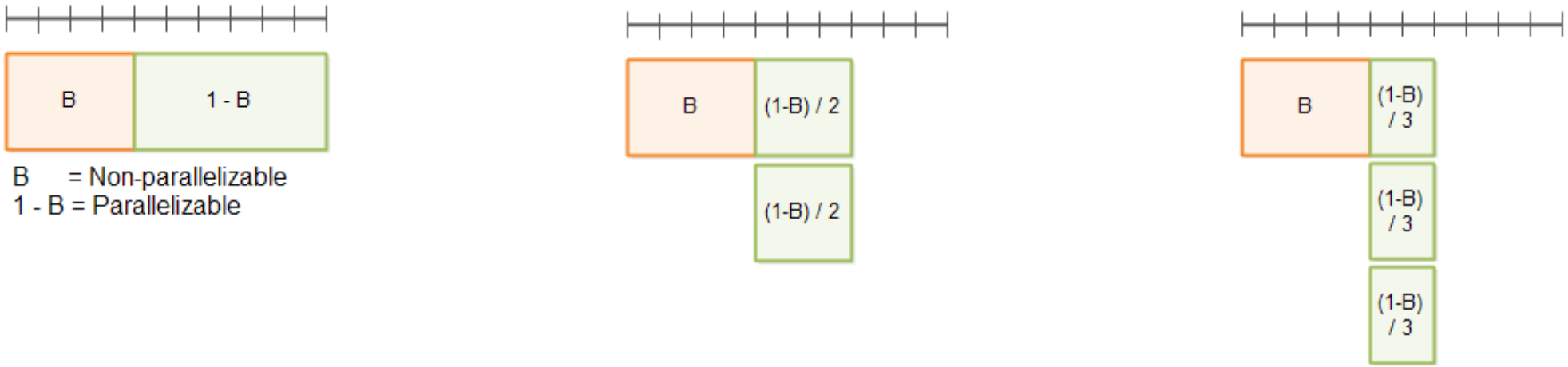
Improvements with concurrency

- Concurrency promises to perform certain tasks faster
  - A task = Several subtasks,
  - These subtasks can be executed in parallel

  → Save time (Better CPU Utilization)
- The theoretical possible performance gain can be calculated by *Amdahl's Law*

Amdahl's Law Illustrated

- If **B** is the percentage of the program which can not run in parallel and **N** is the number of processes, then the maximum performance gain is 1 / (**B** + ((**1-B**)/**N**)).



Saving time

```
5 seconds reading file A
2 seconds processing file A
5 seconds reading file B
2 seconds processing file B
----------------------
14 seconds total
```

```
5 seconds reading file A
5 seconds reading file B + 2 seconds processing file A
2 seconds processing file B
----------------------
12 seconds total
```

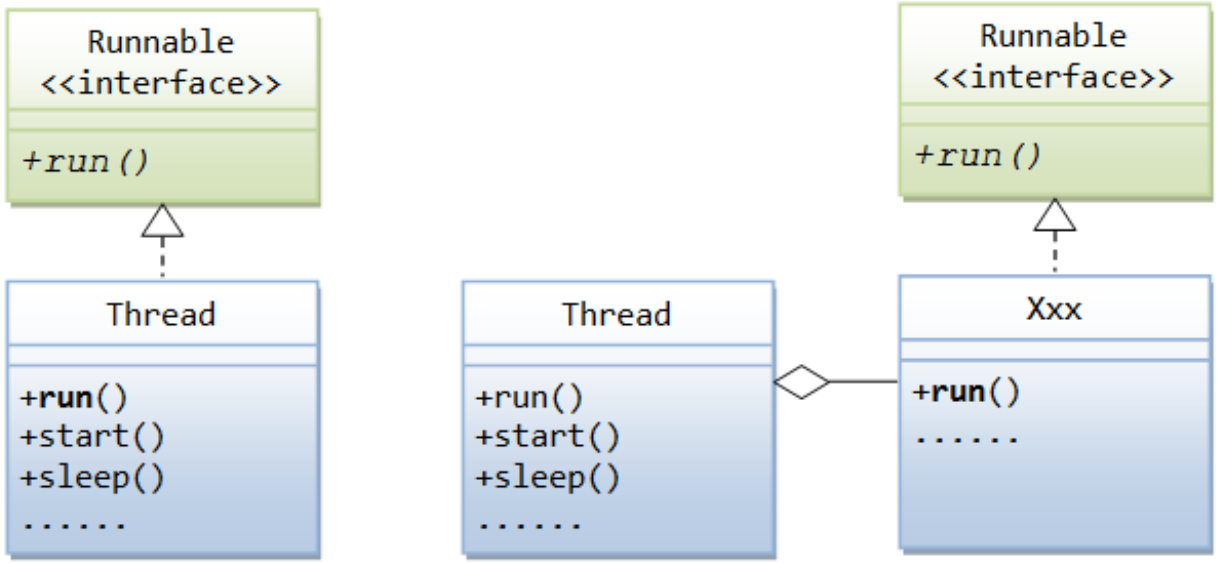
Issues with concurrency

- Threads have their call stack but can also access shared data
- Access problem: if several threads access and change the same shared data at the same time
  - Safety failure (inconsistent data)
  - Thread Interference Errors (Race conditions)
- Visibility problem: if thread A reads shared data, which is later changed by thread B and thread A is unaware of this change
  - Liveness failure (e.g., deadlocks)
  - Memory Consistency Errors

Java Thread

Creating and Starting a Thread

- By extending Thread class
- By providing a Runnable object
  - Runnable Anonymous Class
  - Runnable Lambda Expression



```java
3 public class ThreadExample extends Thread {
4     @Override
5     public void run() {
6         System.out.println("Inside : " + Thread.currentThread().getName());
7     }
8
9     public static void main(String[] args) {
10        System.out.println("Inside : " + Thread.currentThread().getName());
11        Thread thread = new ThreadExample();
12        thread.start();
13    }
14 }
```

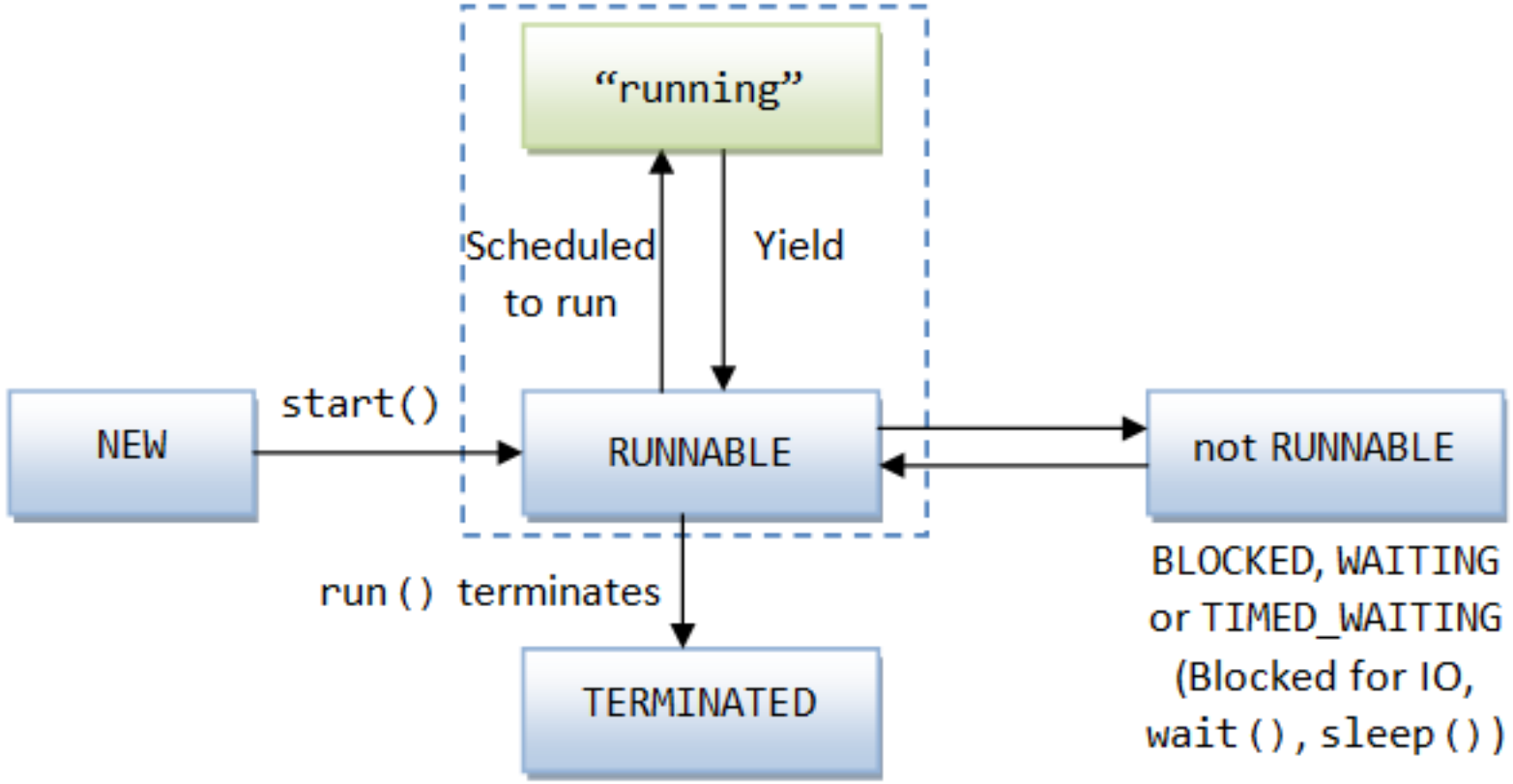
```java
3 public class RunnableExample implements Runnable {
4     @Override
5     public void run() {
6         System.out.println("Inside : " + Thread.currentThread().getName());
7     }
8
9     public static void main(String[] args) {
10        System.out.println("Inside : " + Thread.currentThread().getName());
11        Runnable runnable = new RunnableExample();
12        Thread thread = new Thread(runnable);
13        thread.start();
14    }
15
16 }
```

```java
3 public class RunnableExampleAnonymousClass {
4
5     public static void main(String[] args) {
6         System.out.println("Inside : " + Thread.currentThread().getName());
7         Runnable runnable = new Runnable() {
8             @Override
9             public void run() {
10                 System.out.println("Inside : " + Thread.currentThread().getName());
11             }
12         };
13         Thread thread = new Thread(runnable);
14         thread.start();
15     }
16 }
```

```java
3 public class RunnableExampleLambdaExpression {
4     public static void main(String[] args) {
5         System.out.println("Inside : " + Thread.currentThread().getName());
6         Runnable runnable = () -> {
7             System.out.println("Inside : " + Thread.currentThread().getName());
8         };
9         Thread thread = new Thread(runnable);
10        thread.start();
11    }
12 }
```

Runnable or Thread, which one to use?

The Life Cycle of a Thread



Pausing execution

```java
    public static void main(String[] args) {
        System.out.println("Inside : " + Thread.currentThread().getName());
        String[] messages = {"If I can stop one heart from breaking,",
                "I shall not live in vain.",
                "If I can ease one life the aching,",
                "Or cool one pain,",
                "Or help one fainting robin",
                "Unto his nest again,",
                "I shall not live in vain"};
        Runnable runnable = () -> {
            System.out.println("Inside : " + Thread.currentThread().getName());
            for(String message: messages) {
                System.out.println(message);
                try {
                    Thread.sleep(2000);
                } catch (InterruptedException e) {
                    throw new IllegalStateException(e);
                }
            }
        };
        Thread thread = new Thread(runnable);
        thread.start();
    }
```

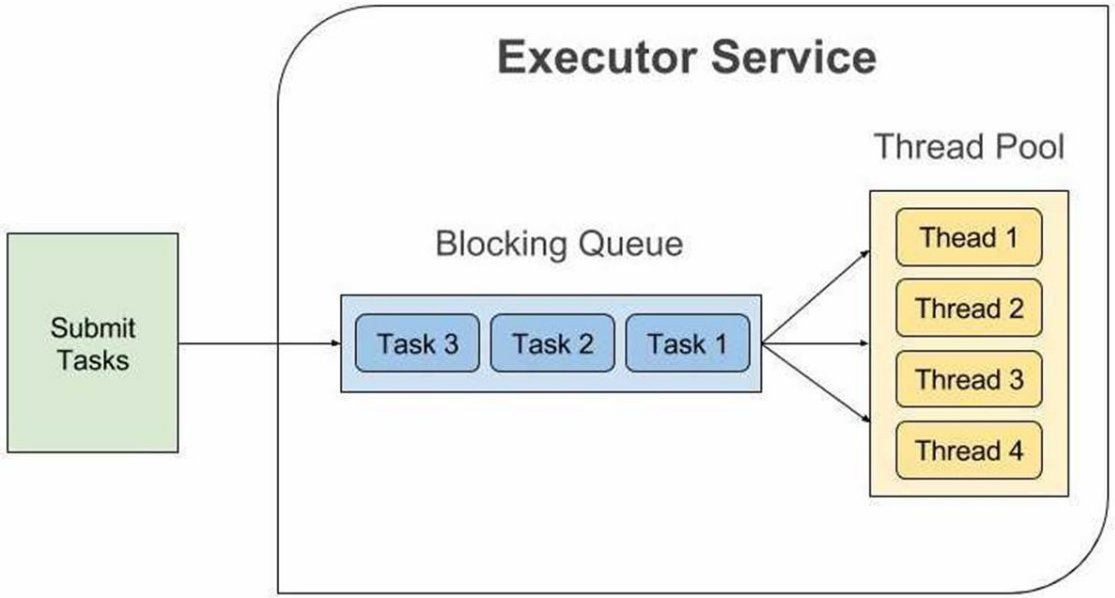
Waiting for completion of another thread

```java
Thread thread1 = new Thread(() -> {
    System.out.println("Entered Thread 1");
    try {
        Thread.sleep(2000);
    } catch (InterruptedException e) {}
    System.out.println("Exiting Thread 1");
});
Thread thread2 = new Thread(() -> {
    System.out.println("Entered Thread 2");
    try {
        Thread.sleep(4000);
    } catch (InterruptedException e) {}
    System.out.println("Exiting Thread 2");
});

System.out.println("Starting Thread 1");
thread1.start();
System.out.println("Waiting for Thread 1 to complete");
try {
    thread1.join(1000);
} catch (InterruptedException e) {}
System.out.println("Waited enough! Starting Thread 2 now");
thread2.start();
```

Executor Framework

Executor Framework

- Based on producer-consumer architecture
- Executor service decouples submission from execution policy to let developers easily specify and modify the implementation later without much of a code change.
- Tasks are submitted to a thread poo
  - If there are more tasks than the numbe for waiting
  - If the queue is full → rejected.



Executor Framework - Functionalities

- **Thread Creation**: Methods for creating threads, a pool of threads, the application can use to run tasks concurrently.
- **Thread Management**: Managing the life cycle of the threads in the thread pool.
  - You don't need to worry about whether the threads in the thread pool are active or busy or dead before submitting a task for execution
- **Task submission and execution**: Methods for submitting tasks for execution in the thread pool and deciding when the tasks will be executed

Java executor interfaces

- **Executor** - A simple interface that contains a method called `execute()` to launch a task specified by a `Runnable` object.
- **ExecutorService** - A sub-interface of `Executor` that adds functionality to manage the lifecycle of the tasks. It also provides a `submit()` method whose overloaded versions can accept a `Runnable` and a `Callable` (discussed later).
- **ScheduledExecutorService** - A sub-interface of `ExecutorService`. It adds functionality to schedule the execution of the tasks.

ExecutorService example

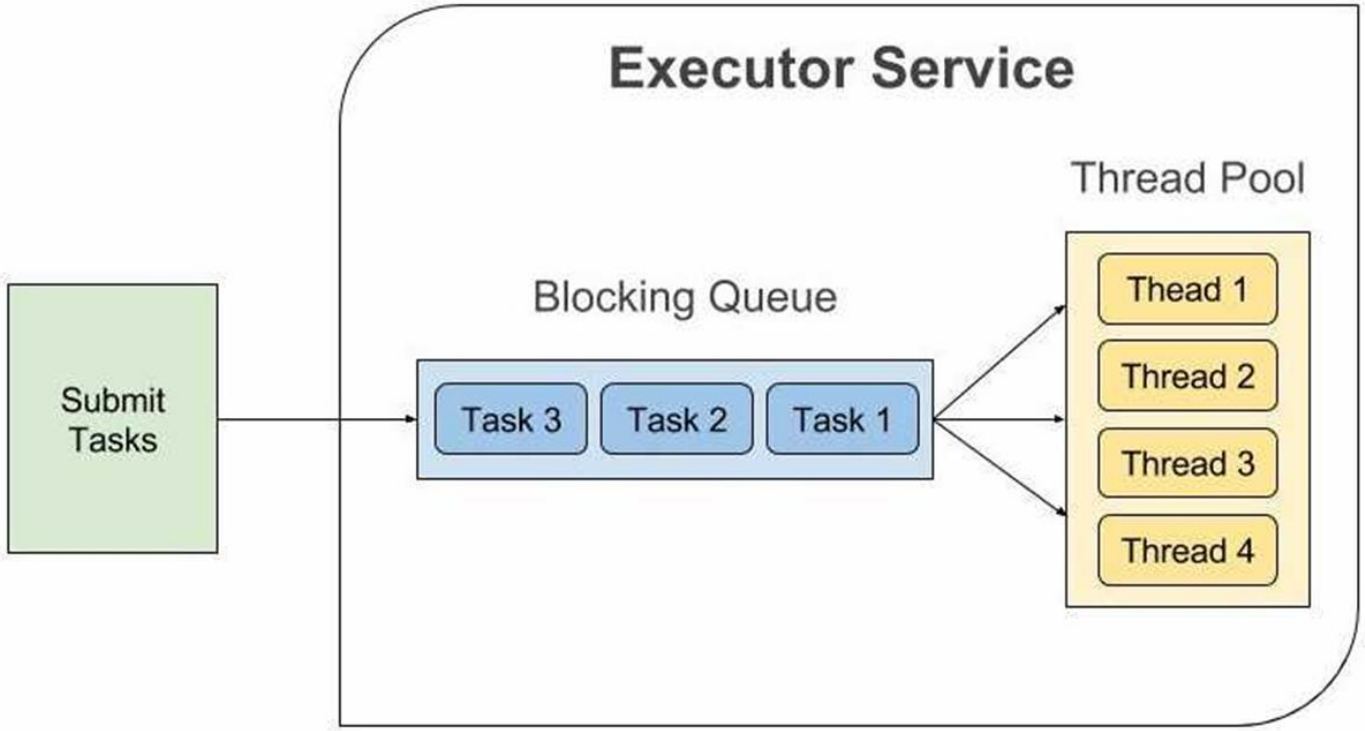
```java
    public static void main(String[] args) {
        System.out.println("Inside : " + Thread.currentThread().getName());

        ExecutorService executorService = Executors.newSingleThreadExecutor();
        Runnable runnable = () -> {
            System.out.println("Inside : " + Thread.currentThread().getName());
        };

        System.out.println("Submit the task specified by the runnable.");
        executorService.submit(runnable);
        executorService.shutdown();
    }
```

```
Inside : main
Submit the task specified by the runnable.
Inside : pool-1-thread-1
```

ExecutorService example with multiple threads and tasks

```
8     public static void main(String[] args) {
9         ExecutorService executorService = Executors.newFixedThreadPool(2);
10        Runnable task1 = () -> {
11            System.out.println("Task1 inside : " + Thread.currentThread().getName());
12            try {
13                TimeUnit.SECONDS.sleep(2);
14            } catch (InterruptedException ex) {}
15        };
16
17        Runnable task2 = () -> {
18            //
19        };
20        Runnable task3 = () -> {
21            //
22        };
23        executorService.submit(task1);
24        executorService.submit(task2);
25        executorService.submit(task3);
26
27        executorService.shutdown();
28    }
```

Thread Pool


Executor Service
Thread Pool
Blocking Queue
Submit Tasks
Task 3
Task 2
Task 1
Thead 1
Thread 2
Thread 3
Thread 4

Callable

Callable

- `Runnable` object to define the tasks that are executed inside a thread

→ What if you want to return a result from your tasks?

→ Java provides a `Callable` interface

- A `Callable` is similar to `Runnable`, except it can return a result and throw a checked exception
- `Callable` interface has a single method `call()` to contain the code executed by a thread

Callable examples

```java
Callable<String> callable = new Callable<String>() {
    @Override
    public String call() throws Exception {
        // Perform some computation
        Thread.sleep(2000);
        return "Return some result";
    }
};
```

```java
//a lambda expression
Callable<String> callable = () -> {
        // Perform some computation
        Thread.sleep(2000);
        return "Return some result";
};
```

Executing Callable tasks

- A `Callable` can be submitted to an executor service for execution
  - What about the `Callable's` result?
  - How do you access it?
  - When the result of the submitted task will be available
- Future can be used to fetch the result of the task when it is available

Future & Callable Example

```java
ExecutorService executorService = Executors.newSingleThreadExecutor();

Callable<String> callable = () -> {
    System.out.println("Entered Callable");
    Thread.sleep(2000);
    return "Hello from Callable";
};

Future<String> future = executorService.submit(callable);

// This line executes immediately
// Do something else while callable is getting executed
// Retrieve the result of the future

// Blocks until the result is available
String result = future.get();
System.out.println(result);

executorService.shutdown();
```

Future & Callable Example

```java
Future<String> future = executorService.submit(() -> {
    Thread.sleep(2000);
    return "Hello from Callable";
});

while(!future.isDone()) {
    System.out.println("Task is still not done...");
    Thread.sleep(200);
}

System.out.println("Task completed! Retrieving the result");
String result = future.get();
System.out.println(result);

executorService.shutdown();
```

Cancelling a Future

- Cancelling a future by using `Future.cancel()` method
- The `cancel()` method accepts `mayInterruptIfRunning`.
  - `True`: the thread that is currently executing the task will be interrupted
  - `Fase`: in-progress tasks will be allowed to complete.
- `isCancelled()`: check if a task is canceled or not.
- After the cancellation of the task, `isDone()` will always be true.

invokeAll & invokeAny

Submit multiple tasks and wait for all of them to complete

- Executing multiple tasks by passing a collection of `Callables` to the `invokeAll()` method
  - `invokeAll()` returns a list of `Futures`
  - Any call to `future.get()` will be blocked until all the `Futures` are complete

Submit multiple tasks and wait for any one of them to complete

- The `invokeAny()` method accepts a collection of `Callables` and returns the result of the fastest `Callable`.

Synchronization

Issues with concurrency

- Threads have their call stack but can also access shared data
- **Access problem**: if several threads access and change the same shared data at the same time
  - Safety failure (inconsistent data)
  - Thread Interference Errors (Race conditions)
- **Visibility problem**: if thread A reads shared data, which is later changed by thread B and thread A is unaware of this change
  - Liveness failure (e.g., deadlocks)
  - Memory Consistency Errors

Output?

```java
ExexecutorService executorService = Executors.newFixedThreadPool(10);

Counter counter = new Counter();

for (int i = 0; i < 1000; i++) {
    executorService.submit(() -> counter.increment());
}

executorService.shutdown();
executorService.awaitTermination(60, TimeUnit.SECONDS);

System.out.println("Final count is : " + counter.getCount());
```

```java
class Counter {
    int count = 0;

    public void increment() {
        count = count + 1;
    }

    public int getCount() {
        return count;
    }
}
```

Final count is : 985

Final count is : 999

Final count is : 997

Output?

```java
private static boolean sayHello = false;
public static void main(String[] args) throws I
    Thread thread = new Thread(() -> {
        while (!sayHello) {
            try {
                Thread.sleep(1000);
            } catch (InterruptedException e) {}
        }
        System.out.println("Hello World!");
        while (sayHello) {
            System.out.println("sayhello is");
        }
        System.out.println("Good Bye!");
    });

    thread.start();

    Thread.sleep(1000);
    System.out.println("Say Hello..");
    sayHello = true;

    Thread.sleep(1000);
    System.out.println("Say Bye..");
    sayHello = false;

}
```

Synchronization

How do we avoid those problems?

1. Only one thread can read and write a shared variable at a time.
   - When one thread is accessing a shared variable, other threads should wait until the first thread is done.
   - This guarantees that the access to a shared variable is `Atomic`, and that multiple threads do not interfere.
2. Whenever any thread modifies a shared variable, it automatically establishes a happens-before relationship with subsequent reads and writes of the shared variable by other threads.
   - This guarantees that changes done by one thread are visible to others.

Synchronization

Synchronized Methods

- The `synchronized` keyword makes sure that only one thread can enter the sync methods at one time

Synchronized Blocks

- Java internally uses a so-called *intrinsic lock or monitor lock* to manage thread synchronization. Every object has an intrinsic lock associated with it.

Synchronized Methods

```java
ExecutorService executorService = Executors.newFixedThreadPool(10);

Counter counter = new Counter();

for (int i = 0; i < 1000; i++) {
    executorService.submit(() -> counter.increment());
}

executorService.shutdown();
executorService.awaitTermination(60, TimeUnit.SECONDS);

System.out.println("Final count is : " + counter.getCount());
```

```java
class Counter {
    int count = 0;

    public synchronized void increment() {
        count = count + 1;
    }

    public int getCount() {
        return count;
    }
}
```

Volatile Keyword

- `Volatile` keyword is used to avoid memory consistency errors in multithreaded programs.
  - It tells the compiler to avoid doing any optimizations to the variable.
  - If you mark a variable as `volatile`, the compiler won't optimize or reorder instructions around that variable.

```
private static volatile boolean sayHello = false;
```

Locks

- `ReentrantLock` is a mutually exclusive lock with the same behavior as the intrinsic/implicit lock accessed via the synchronized keyword.
  - Thread that currently owns the lock can acquire it more than once without any problem.
- The `ReentrantLock` also provides various methods for more fine-grained control
- The `tryLock()` method tries to acquire the lock without pausing the thread.
  - If the thread couldn't acquire the lock because it was held by some other thread, then it returns immediately instead of waiting for the lock to be released.

```java
class ReentrantLockCounter {
    private final ReentrantLock lock = new ReentrantLock();

    private int count = 0;

    // Thread Safe Increment
    public void increment() {
        lock.lock();
        try {
            count = count + 1;
        } finally {
            lock.unlock();
        }
    }
}
```

ReadWriteLock

- `ReadWriteLock` consists of a pair of locks - one for read access and one for write access.
  - The read lock may be held by multiple threads simultaneously as long as the write lock is not held by any thread.
- `ReadWriteLock` allows for an increased level of concurrency.
  - It performs better compared to other locks in applications where there are fewer writes than reads.

```java
ReadWriteLock lock = new ReentrantReadWriteLock();
private int count = 0;

public int incrementAndGetCount() {
    lock.writeLock().lock();
    try {
        count = count + 1;
        return count;
    } finally {
        lock.writeLock().unlock();
    }
}

public int getCount() {
    lock.readLock().lock();
    try {
        return count;
    } finally {
        lock.readLock().unlock();
    }
}
```

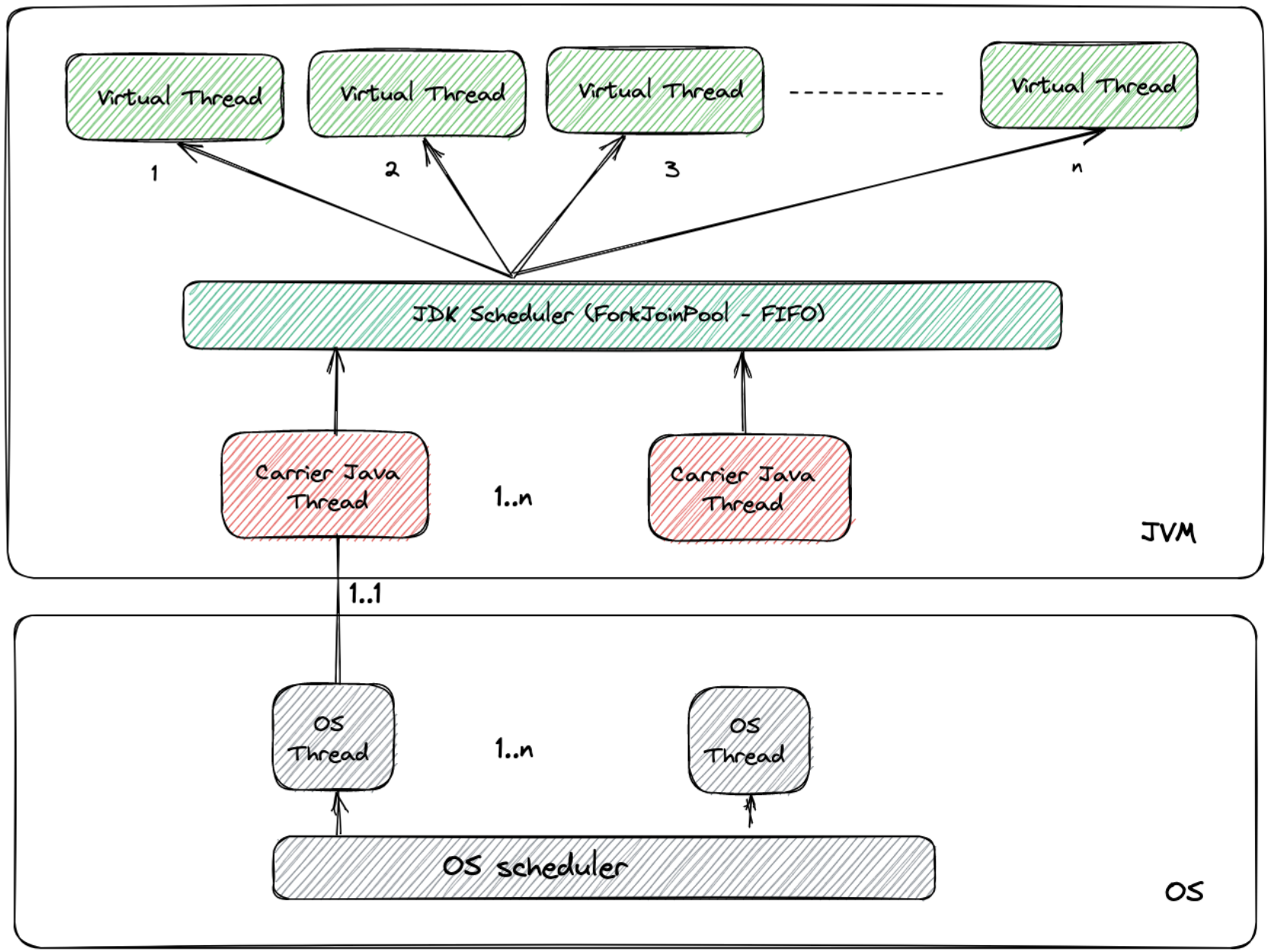
Java Virtual Thread

Java (normal) Thread

- To create a new kernel thread, we must do a system call, and that's a costly operation
  → Using thread pools instead of reallocating and deallocating threads as needed
- Scaling up application by adding more threads
  → Context switching + Memory footprint
  → Cost of thread maintaining may be significant and affect the processing time

Java Virtual Thread

- Virtual threads are managed by the JVM
- Their allocation doesn't require a system call
- They're free of OS's context switch
  - Virtual threads run on *the carrier kernel thread* used under-the-hood
  - More virtual threads…
  - Managing virtual threads is much cheaper


Virtual Thread
Virtual Thread
Virtual Thread
Virtual Thread
1
2
3
n
JDK Scheduler (ForkJoinPool - FIFO)
Carrier Java Thread
1..n
Carrier Java Thread
JVM
1..1
OS Thread
1..n
OS Thread
OS scheduler
OS

API is the same!!!

```java
Runnable printThread = () -> System.out.println(Thread.currentThread());

ThreadFactory virtualThreadFactory = Thread.builder().virtual().factory();
ThreadFactory kernelThreadFactory = Thread.builder().factory();

Thread virtualThread = virtualThreadFactory.newThread(printThread);
Thread kernelThread = kernelThreadFactory.newThread(printThread);

virtualThread.start();
kernelThread.start();
```